

 l^p -boundedness properties for Volterra difference equationsClaudio Cuevas^{a,*}, Filipe Dantas^a, Mario Choquehuanca^b, Herme Soto^b^aDepartamento de Matemática, Universidade Federal de Pernambuco Recife-PE, CEP. 50540-740, Brazil^bDepartamento de Matemática y Estadística, Universidad de La Frontera, Casilla 54-D Temuco, Chile

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ABSTRACT

This work deals with l^p -boundedness of solutions of a class of Volterra functional difference equation. The effect of different kinds of the nonhomogeneous term (Lipschitz, local Lipschitz, completely continuous), are taken into consideration.

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1. Introduction and statement of results

Let $\mathbb{X}$ be an arbitrary Banach space. In this work, we study l^p -boundedness properties¹ for Volterra functional difference equation given by

$$u(n+1) = \lambda \sum_{j=-\infty}^n a(n-j)u(j) + f(n, u_n), \quad n \in \mathbb{Z}, \quad (1.1)$$

where λ is a complex number, $a(n)$ is a summable $\mathbb{C}$ -valued function, f is an appropriate function and $u_n : \mathbb{Z}_- \rightarrow \mathbb{X}$ is the pre-history function, which is defined by $u_n(\theta) = u(n+\theta)$ for all $\theta \in \mathbb{Z}_-$.

Until now the literature concerning discrete l^p -properties for functional difference equations is too incipient and should be extensively studied. For example in [3] the relation between the following two properties of linear difference equations with infinite delay is investigated: (i) exponential stability and (ii) l^p -input l^q -state stability (Perron's property). The main result of them state that (i) $\Leftrightarrow$ (ii) whenever $(p, q) \neq (1, \infty)$ and a certain boundedness condition on coefficients is fulfilled. In [1] the authors have studied existence of bounded solutions which are in l^p for semilinear functional difference equations with infinite delay. To get the result they have used exponential dichotomy notion and a new approach based on discrete maximal regularity (see [7]). We observe that abstract retarded functional difference equations describe processes, whose current state is determined by their entire prehistory. These processes are encountered for example in mathematical models in population dynamics, as well as in models of propagation of perturbation in materials with memory, various models to describe the evolution of epidemics, the theory of viscoelasticity and the study of optimal control problems (cf. [12,20–23] and references therein).

Properties of the solutions for Volterra functional difference equations have been studied in several contexts. For example, invariant manifolds theory [24], convergence theory [8,10], discrete maximal regularity [9], asymptotic behavior [7,25], exponential dichotomy and robustness [4], stability [3,14,15], periodicity [2,5,11,16,28]. An extensive list of applications can be found in [21].

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¹ The study of l^p -boundedness properties for operators is very important in classical analysis and ODEs. Many discrete operators present distinctive difficulties apparently unapproachable from the continuous setting. The study of discrete analogs in harmonic analysis originated in the early 1900's. But it was not until the late 1980's that a renewed study of discrete analogs began in earnest ([29,19]).

In [5] the authors began a study of the existence of discrete almost automorphic solutions of Volterra functional difference equations of type (1.1) in the case when the perturbation f satisfies a global Lipschitz condition. Now we continue such work from the point of view of ℓ^p -boundedness properties of (1.1).

1.1. ℓ^p -boundedness properties

In this subsection we consider the question of existence and uniqueness of bounded solutions for (1.1) which are in ℓ^p ($p \geq 1$) under several type of perturbations. We now describe the main results which are not known in the literature. The results are interesting because they can be easily used to obtain information about ℓ^p -boundedness of solutions for a large class of equations.

Theorem 1.1. Assume that $\mathcal{B}$ is a phase space² that satisfies axiom (B). Let λ be in Ω_s (see Section 2.2) and let $f : \mathbb{Z} \times \mathcal{B} \rightarrow \mathbb{X}$ be a function that satisfies the following Lipschitz condition:

$$\|f(k, \varphi) - f(k, \psi)\| \leq L_f(k) \|\varphi(0) - \psi(0)\|, \quad (1.2)$$

for all $\varphi, \psi \in \mathcal{B}$ and each $k \in \mathbb{Z}$, where $L_f : \mathbb{Z} \rightarrow \mathbb{R}^+$ is ℓ^p -summable and $f(\cdot, 0) \in \ell^p(\mathbb{Z}, \mathbb{X})$. Then there is a unique bounded solution $u(n)$ of Eq. (1.1) such that $u \in \ell^p(\mathbb{Z}, \mathbb{X})$.

Next we establish a version of Theorem 1.1 which enable us to consider locally Lipschitz perturbations of Eq. (1.1). In the process of obtaining this result we require the assumption:

Condition ($\mathbf{H}_{\text{loc}}$): The following conditions hold:

- (i) The function $f(k, \varphi)$ is locally Lipschitz with respect to $\varphi \in \mathcal{B}$, i.e. for each positive number R , for all $k \in \mathbb{Z}$ and $\|\varphi\|_{\mathcal{B}} \leq R$, $\|\psi\|_{\mathcal{B}} \leq R$

$$\|f(k, \varphi) - f(k, \psi)\| \leq \ell(k, R) \|\varphi(0) - \psi(0)\|,$$

where $\ell : \mathbb{Z} \times [0, \infty) \rightarrow [0, \infty)$ is a nondecreasing function with respect to the second variable.

- (ii) There is a positive number a such that $\sum_{j=-\infty}^{\infty} \ell(j, a)^p < +\infty$.
- (iii) $f(\cdot, 0) \in \ell^p(\mathbb{Z}, \mathbb{X})$.

We have the following local version of Theorem 1.1.

Theorem 1.2. Let λ be in Ω_s . Assume that $\mathcal{B}$ is a phase space that satisfies axiom (B) and that the condition ($\mathbf{H}_{\text{loc}}$) holds. Then, there is a constant $m \in \mathbb{Z}$ and a unique ℓ^p -bounded solution $u(n)$ of (1.1) for all $n \leq m$ such that $u(n) = 0$ for all $n \geq m + 1$ and $\|u\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq a$, where a is the constant of condition (ii) in ($\mathbf{H}_{\text{loc}}$).

The proof of previous theorems relies on the contraction principle.³ To understand more general sort of perturbations, we need to deal with more general fixed point theorems than contraction principle. A investigation in such direction is justified not only by the aspiration for mathematical generality, but also by the fact that the overwhelming majority of real problems are nonlinear.

To obtain our next result, we require that the following condition holds.

Condition ($\mathbf{H}_{\text{sca}}$): Let $f : \mathbb{Z} \times \mathcal{B} \rightarrow \mathbb{C}$ be a function such that the following conditions hold:

- (i) There is a function $\mathcal{W}_f : \mathbb{Z} \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ nondecreasing with respect to the second variable with

$$\sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, r)^p < \infty,$$

for all $r \geq 0$ so that $|f(k, \varphi)| \leq \mathcal{W}_f(k, \|\varphi\|_{\mathcal{B}})$, for all $k \in \mathbb{Z}$ and $\varphi \in \mathcal{B}$.

- (ii) The map $\varphi \in \mathcal{B} \rightarrow f(\cdot, \varphi) \in \ell^p(\mathbb{Z}, \mathbb{C})$ is uniformly continuous.
- (iii) There is $R > 0$ such that $\frac{1}{R} \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, \rho R)^p \right)^{1/p} < 1$.

We obtain the following result.⁴

Theorem 1.3. Let λ be in Ω_s . Assume that $\mathcal{B}$ is a scalar phase space that satisfies axiom (B) and that the condition ($\mathbf{H}_{\text{sca}}$) holds. Then, there is a ℓ^p -bounded solution $u(n)$ of (1.1).

² Here we use fairly the notations and terminology for phase space. See Hino et al. [18] for an outline of the general philosophy of such spaces. The idea of considering phase spaces for studying qualitative properties of functional difference equations was used first by Murakami [26] for the study of some spectral properties of the solution operator for linear Volterra difference systems and then by Elaydi et al. [13] for the study of asymptotic equivalence of bounded solutions of a homogeneous Volterra difference system and its perturbation.

³ The usefulness of the fixed point methods for applications have increased enormously by the development of accurate and efficient techniques for computing fixed point, making fixed point arguments a powerful weapon in the arsenal of applied mathematics (see [17]).

⁴ The framework for the proof of Theorem 1.3 uses the Schauder's fixed point theorem and a compactness criterion in $\ell^p(\mathbb{Z}, \mathbb{C})$ (see [30, p.286]).

We will now present a summary of this work. Section 2 provides the definitions and preliminary results to be used in the theorems stated and proved in this article. In particular, to facilitate access to the individual topics, we review in Section 2.1 the phase space axioms while in Section 2.2, we present the notion of fundamental solution associated to the Eq. (1.1). In Section 3, we prove Theorems 1.1, 1.2 and 1.3. Finally in Section 4 our methods are applied to other equations.

2. Preliminaries and basic results

It is convenient to introduce some notations. We denote by $\mathbb{R}, \mathbb{C}, \mathbb{Z}, \mathbb{Z}_-$ and $\mathbb{Z}_+$ the set of all real numbers, the set of complex numbers, the set of all integers, the set of all non-positive integers and the set of all non-negative integers, respectively. Fix once and all a complex Banach space $(\mathbb{X}, \|\cdot\|)$. We denote $B_R(\mathbb{X})$ the closed ball in $\mathbb{X}$ with center 0 and radius R .

2.1. Phase space axioms

To study functional difference equations with infinite delay we need to introduce the concept of phase space. In this paper, we use the axiomatic definition of phase space $\mathcal{B}$ following [8,25,27]. Thus $\mathcal{B}$ will denote a vector space of functions defined from $\mathbb{Z}_-$ into $\mathbb{X}$ endowed with a norm denoted $\|\cdot\|_{\mathcal{B}}$ so that $(\mathcal{B}, \|\cdot\|_{\mathcal{B}})$ is a Banach space and the following axiom holds:

(A) There are a positive constant J and nonnegative functions $K(\cdot), M(\cdot)$ defined on $\mathbb{Z}_-$ having the following property: If $x: \mathbb{Z} \rightarrow \mathbb{X}$ is a function such that $x_0 \in \mathcal{B}$, then for all $n \in \mathbb{Z}_+$ the following conditions are fulfilled:

- (i) $x_n \in \mathcal{B}$,
- (ii) $J\|x(n)\| \leq H\|x_n\|_{\mathcal{B}} \leq K(n)\max_{0 \leq i \leq n}\|x(i)\| + M(n)\|x_0\|_{\mathcal{B}}$.

To obtain our results, we consider also the following axiom:

(B) If $(\varphi^n)_{n \in \mathbb{N}}$ is a uniformly bounded sequence in $\mathcal{B}$ which converges pointwise to φ , then $\varphi \in \mathcal{B}$ and $\|\varphi^n - \varphi\|_{\mathcal{B}} \rightarrow 0$ as $n \rightarrow \infty$.

Remark 2.1. The notation $B(\mathbb{Z}_-, \mathbb{X})$ stands for the space consisting of bounded functions from $\mathbb{Z}_-$ into $\mathbb{X}$ endowed with the norm $\|\cdot\|_{\infty}$ of uniform convergence. It is clear that if axiom (B) holds, then $B(\mathbb{Z}_-, \mathbb{X})$ is continuously included in $\mathcal{B}$. Throughout the rest of the paper, $\rho > 0$ denotes a constant such that $\|\varphi\|_{\mathcal{B}} \leq \rho\|\varphi\|_{\infty}$ for every $\varphi \in B(\mathbb{Z}_-, \mathbb{X})$.

Example 2.1. Let $\gamma > 0$. We define $\mathcal{B}^{\gamma}(\mathbb{X})$ as the space consisting of all functions $\varphi: \mathbb{Z}_- \rightarrow \mathbb{X}$ such that $\sup_{i \in \mathbb{Z}_-} \|\varphi(i)\|e^{\gamma i} < \infty$ endowed with the norm

$$\|\varphi\|_{\gamma} = \sup_{i \in \mathbb{Z}_-} \|\varphi(i)\|e^{\gamma i}.$$

It is well known [27] that this space satisfies axiom (A). Moreover, if $(\varphi^n)_{n \in \mathbb{N}}$ is a uniformly bounded sequence in $\mathcal{B}^{\gamma}(\mathbb{X})$ and $\varphi^n(i) \rightarrow \varphi(i)$ as $n \rightarrow \infty$ for all $i \in \mathbb{Z}_-$, then φ is bounded, so that $\varphi \in \mathcal{B}^{\gamma}(\mathbb{X})$ and $\|\varphi^n - \varphi\|_{\gamma} \rightarrow 0$ as $n \rightarrow \infty$.

Example 2.2. Let $1 \leq p < \infty$ and let $g: \mathbb{Z}_- \rightarrow \mathbb{R}$ be a positive function such that $\sum_{i=-\infty}^0 g(i) < \infty$ and $g(0) > 0$. We define $S_g^p(\mathbb{X})$ as the space consisting of all functions $\varphi: \mathbb{Z}_- \rightarrow \mathbb{X}$ such that $\sum_{i=-\infty}^0 \|\varphi(i)\|^p g(i) < \infty$, endowed with the norm

$$\|\varphi\|_{S_g^p} = \left(\sum_{i=-\infty}^0 \|\varphi(i)\|^p g(i) \right)^{1/p}.$$

It is easy to note that this space satisfies (A). Moreover, if $(\varphi^n)_{n \in \mathbb{N}}$ is a uniformly bounded sequence in $S_g^p(\mathbb{X})$ and $\varphi^n(i) \rightarrow \varphi(i)$ as $n \rightarrow \infty$ for all $i \in \mathbb{Z}_-$, then $\varphi \in S_g^p(\mathbb{X})$ and $\|\varphi^n - \varphi\|_{S_g^p} \rightarrow 0$ as $n \rightarrow \infty$.

2.2. Linear Volterra difference equations

In this section we are concerned with the study of the existence of l^p -solutions for linear Volterra difference equations described by

$$u(n+1) = \lambda \sum_{j=-\infty}^n a(n-j)u(j) + f(n), \quad n \in \mathbb{Z}, \quad (2.1)$$

where λ is a complex number, $a: \mathbb{N} \rightarrow \mathbb{C}$ is a summable function and f is in $l^p(\mathbb{Z}, \mathbb{X})$. For a given $\lambda \in \mathbb{C}$, let $s(\lambda, k) \in \mathbb{C}$ be the solution of the difference equation

$$s(\lambda, k+1) = \lambda \sum_{j=0}^k a(k-j)s(\lambda, j), \quad k = 0, 1, 2, \dots, \quad s(\lambda, 0) = 1. \quad (2.2)$$

In this case, $s(\lambda, k)$ is called the fundamental solution to the equation (2.1) generated by $a(\cdot)$. We define the set $\Omega_s := \{\lambda \in \mathbb{C} : \|s(\lambda, \cdot)\|_1 := \sum_{k=0}^{\infty} |s(\lambda, k)| < +\infty\}$.

Example 2.3. For $a(k) = \varsigma^k$, where $|\varsigma| < 1$, we obtain, after a calculation using in (2.2) the unilateral Z-transform, that $s(\lambda, k) = \lambda(\lambda + \varsigma)^{k-1}$, $k \geq 1$ and hence $\mathbb{D}(-\varsigma, 1) := \{z \in \mathbb{C} : |z + \varsigma| < 1\} \subseteq \Omega_\varsigma$.

The next result was inspired in a result about maximal regularity in [7, Theorem 3.1].

Theorem 2.1. Let λ be in Ω_ς . Then for any $f \in l^p(\mathbb{Z}, \mathbb{X})$ the equation (2.1) has a unique solution $u(n)$ in $l^p(\mathbb{Z}, \mathbb{X})$ which is given by

$$u(n+1) = \sum_{j=-\infty}^n s(\lambda, n-j)f(j). \quad (2.3)$$

The solution $u(n)$ satisfies $u \in l^{p'}(\mathbb{Z}, \mathbb{X})$ for all $1 \leq p \leq p' \leq \infty$ and the following estimate holds:

$$\|u\|_{l^{p'}(\mathbb{Z}, \mathbb{X})} \leq \|s(\lambda, \cdot)\|_1^{1-\frac{1}{p}+\frac{1}{p'}} \|s(\lambda, \cdot)\|_\infty^{\frac{1}{p}} \|f\|_{l^p(\mathbb{Z}, \mathbb{X})}. \quad (2.4)$$

In particular, if $p = \infty$, we get

$$\|u\|_\infty \leq \|s(\lambda, \cdot)\|_1 \|f\|_\infty. \quad (2.5)$$

Proof. We divide the proof into several steps.

Step 1 Let $u(n)$ be the function given by (2.3). Taking into account that $a(\cdot)$ is a summable function, we obtain that

$$\begin{aligned} \sum_{j=-\infty}^n a(n-j)\lambda u(j) &= \lambda \sum_{j=-\infty}^n a(n-j) \left(\sum_{\tau=-\infty}^{j-1} s(\lambda, j-1-\tau)f(\tau) \right) = \lambda \sum_{j=-\infty}^{n-1} \sum_{\tau=-\infty}^j a(n-1-j)s(\lambda, j-\tau)f(\tau) \\ &= \lambda \sum_{\tau=-\infty}^{n-1} \sum_{j=\tau}^{n-1} a(n-1-j)s(\lambda, j-\tau)f(\tau) = \sum_{\tau=-\infty}^{n-1} \left(\sum_{j=0}^{n-1-\tau} \lambda a(n-1-\tau-j)s(\lambda, j) \right) f(\tau) = \sum_{\tau=-\infty}^{n-1} s(\lambda, n-\tau)f(\tau) \\ &= \sum_{\tau=-\infty}^n s(\lambda, n-\tau)f(\tau) - s(\lambda, 0)f(n) = u(n+1) - f(n), \end{aligned}$$

which establishes that $u(\cdot)$ is the solution of Eq. (2.1).

Step 2 Let p and q be conjugated exponents. From (2.3) and applying Hölder's inequality we have the following estimate:

$$\|u(n)\|^{p'} \leq \|s(\lambda, \cdot)\|_1^{\frac{p'}{q}} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| |f(j)|^p \right)^{\frac{p'}{p}} \leq \|s(\lambda, \cdot)\|_1^{\frac{p'}{q}} \|s(\lambda, \cdot)\|_\infty^{p'-p} \|f\|_{l^p(\mathbb{Z}, \mathbb{X})}^{p'-p} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| |f(j)|^p \right).$$

Then

$$\|u\|_{l^{p'}(\mathbb{Z}, \mathbb{X})}^{p'} \leq \|s(\lambda, \cdot)\|_1^{\frac{p'}{q}} \|s(\lambda, \cdot)\|_\infty^{p'-p} \|f\|_{l^p(\mathbb{Z}, \mathbb{X})}^{p'-p} \|s(\lambda, \cdot)\|_1 \left(\sum_{n=-\infty}^{\infty} \|f(n)\|^p \right).$$

Therefore

$$\|u\|_{l^{p'}(\mathbb{Z}, \mathbb{X})} \leq \|s(\lambda, \cdot)\|_1^{\frac{1}{q}+\frac{1}{p'}} \|s(\lambda, \cdot)\|_\infty^{\frac{1}{p}} \|f\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

This leads to the desired estimate (2.4).

Step 3 The uniqueness it follows from the following argument: all bounded solutions of (2.1) is given by (2.3). In fact, let $u(n)$ be a bounded function satisfying (2.1), we obtain

$$\begin{aligned} \sum_{j=-\infty}^n s(\lambda, n-j)f(j) &= \sum_{j=-\infty}^n s(\lambda, n-j) \left[u(j+1) - \lambda \sum_{l=-\infty}^j a(j-l)u(l) \right] = \sum_{j=-\infty}^n s(\lambda, n-j)u(j+1) - \lambda \sum_{j=-\infty}^n \sum_{l=-\infty}^j a(j-l)s(\lambda, n-j)u(l) \\ &= \sum_{l=-\infty}^{n+1} s(\lambda, n+1-l)u(l) - \lambda \sum_{l=-\infty}^n \sum_{j=l}^n a(j-l)s(\lambda, n-j)u(l) \\ &= \sum_{l=-\infty}^{n+1} s(\lambda, n+1-l)u(l) - \sum_{l=-\infty}^n \left(\lambda \sum_{j=0}^{n-l} a(n-l-j)s(\lambda, j) \right) u(l) \\ &= \sum_{l=-\infty}^{n+1} s(\lambda, n+1-l)u(l) - \sum_{l=-\infty}^n s(\lambda, n+1-l)u(l) = u(n+1). \end{aligned}$$

Hence, $u(n)$ satisfies (2.3). This completes the proof of Theorem 2.1. $\square$

3. l^p -solutions

In this section we prove [Theorems 1.1, 1.2 and 1.3](#). We begin with the following technical property which will be useful in the proof of [Theorem 1.1](#).

Proposition 3.1. *Let $(\alpha_n)_{n \in \mathbb{Z}}$ be a summable sequence of positive numbers. For all $l \in \mathbb{Z}_+$, one has*

$$\sum_{m=-\infty}^{\infty} \alpha_m \left(\sum_{\tau=-\infty}^{m-1} \alpha_\tau \right)^l \leq \frac{1}{l+1} \left(\sum_{m=-\infty}^{\infty} \alpha_m \right)^{l+1}. \quad (3.1)$$

Proof. Putting $A_m := \sum_{\tau=-\infty}^{m-1} \alpha_\tau$. Following Cuevas and Lizama [\[6, Lemma 3.2\]](#) we obtain

$$\begin{aligned} (l+1)(A_{m+1} - A_m)A_m^l &= (A_{m+1} - A_m)(A_m^l + A_m^{l-1}A_m + \dots + A_mA_m^{l-1} + A_m^l) \\ &\leq (A_{m+1} - A_m)(A_{m+1}^l + A_{m+1}^{l-1}A_m + \dots + A_{m+1}A_m^{l-1} + A_m^l) = A_{m+1}^{l+1} - A_m^{l+1}. \end{aligned}$$

For $n, j \in \mathbb{Z}^+$, we have that

$$(l+1) \sum_{m=-j}^{n-1} (A_{m+1} - A_m)A_m^l \leq \sum_{m=-j}^{n-1} (A_{m+1}^{l+1} - A_m^{l+1}) = A_n^{l+1} - A_{-j}^{l+1}.$$

Letting $n, j \rightarrow \infty$, the inequality [\(3.1\)](#) follows. $\square$

3.1. Proof of Theorem 1.1

Proof. of [Theorem 1.1](#) We define the operator $\mathfrak{F}: l^p(\mathbb{Z}, \mathbb{X}) \rightarrow l^p(\mathbb{Z}, \mathbb{X})$ by

$$\mathfrak{F}u(n) = \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)f(j, u_j). \quad (3.2)$$

We observe that if u is in $l^p(\mathbb{Z}, \mathbb{X})$, then $u \in l^\infty(\mathbb{Z}, \mathbb{X})$. Hence $u_j \in \mathcal{B}$ for each $j \in \mathbb{Z}$.

Let p and q be conjugated exponents. We have the following estimates:

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \|\mathfrak{F}u(n)\|^p &= \sum_{n=-\infty}^{\infty} \left\| \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)f(j, u_j) \right\|^p \leq \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| (\|f(j, u_j) - f(j, 0)\| + \|f(j, 0)\|) \right)^p \\ &\leq \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| L_f(j) \|u(j)\| + \sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, 0)\| \right)^p \\ &\leq 2^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| L_f(j) \|u(j)\| \right)^p + 2^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, 0)\| \right)^p \\ &\leq 2^p \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right)^{\frac{p}{q}} \sum_{n=-\infty}^{\infty} \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right) L_f(n)^p \|u(n)\|^p + 2^p \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right)^{\frac{p}{q}} \sum_{n=-\infty}^{\infty} \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right) \|f(n, 0)\|^p. \quad (3.3) \end{aligned}$$

Therefore we have the upper bound

$$\|\mathfrak{F}u\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq 2 \|s(\lambda, \cdot)\|_1 \left(\|L_f\|_{l^p(\mathbb{Z})} \|u\|_{l^p(\mathbb{Z}, \mathbb{X})} + \|f(\cdot, 0)\|_{l^p(\mathbb{Z}, \mathbb{X})} \right).$$

Hence $\mathfrak{F}$ is well defined.

We now show that the operator $\mathfrak{F}$ has a unique fixed point. Let u and v be two functions in $l^p(\mathbb{Z}, \mathbb{X})$. In view of condition [\(1.2\)](#) we have that

$$\begin{aligned} \|\mathfrak{F}u - \mathfrak{F}v\|_{l^p(\mathbb{Z}, \mathbb{X})} &= \left(\sum_{n=-\infty}^{\infty} \left\| \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)(f(j, u_j) - f(j, v_j)) \right\|^p \right)^{\frac{1}{p}} \leq \left(\sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| L_f(j) \|u(j) - v(j)\| \right)^p \right)^{\frac{1}{p}} \\ &\leq \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right)^{\frac{1}{q}} \left(\sum_{n=-\infty}^{\infty} \left(\sum_{j=0}^{\infty} |s(\lambda, j)| \right) L_f(n)^p \|u(n) - v(n)\|^p \right)^{\frac{1}{p}} \leq \|s(\lambda, \cdot)\|_1 \|L_f\|_{l^p(\mathbb{Z})} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}. \end{aligned}$$

Next, we consider the iterates of the operator $\mathfrak{F}$. We are going next to estimate the l^p -norm of the difference $\mathfrak{F}^{(m)}u - \mathfrak{F}^{(m)}v$. We have initially the following estimate:

$$\|\mathfrak{F}u(n) - \mathfrak{F}v(n)\| \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \|u(j) - v(j)\|^p \right)^{\frac{1}{p}} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}, \quad (3.4)$$

whence

$$\|\mathfrak{F}u(n) - \mathfrak{F}v(n)\|^p \leq \|s(\lambda, \cdot)\|_1^p \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right) \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}^p. \quad (3.5)$$

Using (3.5) and Proposition 3.1, we obtain

$$\begin{aligned} \|\mathfrak{F}^{(2)}u - \mathfrak{F}^{(2)}v\|_{l^p(\mathbb{Z}, \mathbb{X})} &\leq \|s(\lambda, \cdot)\|_1 \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \|\mathfrak{F}u(n) - \mathfrak{F}v(n)\|^p \right)^{\frac{1}{p}} \\ &\leq \|s(\lambda, \cdot)\|_1 \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \|s(\lambda, \cdot)\|_1^p \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right) \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}^p \right)^{\frac{1}{p}} \\ &\leq \|s(\lambda, \cdot)\|_1^2 \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right) \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \left(\frac{a^2}{2} \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}, \end{aligned}$$

where $a := \|s(\lambda, \cdot)\|_1^p \|L_f\|_{l^p(\mathbb{Z})}^p$. Now, using estimates (3.4) and (3.5), we obtain

$$\begin{aligned} \|\mathfrak{F}^{(2)}u(n) - \mathfrak{F}^{(2)}v(n)\| &\leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \|\mathfrak{F}u(j) - \mathfrak{F}v(j)\|^p \right)^{\frac{1}{p}} \\ &\leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \|s(\lambda, \cdot)\|_1^p \left(\sum_{\tau=-\infty}^{j-1} L_f(\tau)^p \right) \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}^p \right)^{\frac{1}{p}} \\ &\leq \|s(\lambda, \cdot)\|_1^2 \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \left(\sum_{\tau=-\infty}^{j-1} L_f(\tau)^p \right) \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})} \\ &\leq \frac{\|s(\lambda, \cdot)\|_1^2}{2^{\frac{1}{p}}} \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right)^{\frac{2}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}. \end{aligned} \quad (3.6)$$

We have in a position to calculate the iterate $\mathfrak{F}^{(3)}$. Using (3.6) and Proposition 3.1 we get

$$\begin{aligned} \|\mathfrak{F}^{(3)}u - \mathfrak{F}^{(3)}v\|_{l^p(\mathbb{Z}, \mathbb{X})} &\leq \|s(\lambda, \cdot)\|_1 \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \|\mathfrak{F}^{(2)}u(n) - \mathfrak{F}^{(2)}v(n)\|^p \right)^{\frac{1}{p}} \\ &\leq \frac{\|s(\lambda, \cdot)\|_1^3}{2^{\frac{1}{p}}} \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right)^2 \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})} \\ &\leq \|s(\lambda, \cdot)\|_1^3 \left(\frac{1}{6} \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \right)^3 \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \left(\frac{a^3}{3!} \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}. \end{aligned}$$

Finally, one must to deal with the iterate $\mathfrak{F}^{(m)}$. An induction argument show us that:

$$\|\mathfrak{F}^{(m-1)}u(n) - \mathfrak{F}^{(m-1)}v(n)\| \leq \frac{\|s(\lambda, \cdot)\|_1^{m-1}}{[(m-1)!]^{\frac{1}{p}}} \left(\sum_{j=-\infty}^{n-1} L_f(j)^p \right)^{\frac{m-1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

Consequently, we get

$$\|\mathfrak{F}^{(m)}u - \mathfrak{F}^{(m)}v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \left(\frac{a^m}{m!} \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

Bearing in mind that $\left(\frac{a^m}{m!} \right)^{\frac{1}{p}} < 1$ holds for n sufficiently large, by the fixed point iteration method $\mathfrak{F}$ has a unique fixed point $u \in l^p(\mathbb{Z}, \mathbb{X})$. This ends the proof of the theorem. $\square$

Example 3.1. Let $\gamma > 0$. We consider the phase space $\mathcal{B}^\gamma(\mathbb{X})$ defined in Example 2.1. Let ς be in $\mathbb{C}$ such that $|\varsigma| < 1$ and take $\lambda \in \mathbb{D}(-\varsigma, 1)$. Let $l: \mathbb{Z} \rightarrow \mathbb{R}^+$ be a function such that $\sum_{j=-\infty}^{\infty} l(j)^p < +\infty$. We consider the following difference equation in the Banach space $\mathbb{X}$:

$$u(n+1) = \lambda \sum_{j=-\infty}^n \varsigma^{n-j} u(j) + \frac{l(n)u(n)}{\|u(n)\| + 1}, \quad n \in \mathbb{Z}. \quad (3.7)$$

Let $f(k, \varphi)$ be the perturbation associated to (3.7), we get the following estimate:

$$\|f(k, \varphi) - f(k, \psi)\| \leq 3l(k)\|\varphi(0) - \psi(0)\|,$$

for all $\varphi, \psi \in \mathcal{B}^\gamma(\mathbb{X})$ and $k \in \mathbb{Z}$. By Theorem 1.1 there is a unique l^p -bounded solution of (3.7).

Corollary 3.1. Let μ be in $\mathbb{Z}_+$ and λ be in Ω_s . Assume that $\mathcal{B}$ is a phase space that satisfies axiom (B). Let $f: \mathbb{Z} \times \mathcal{B} \rightarrow \mathbb{X}$ be a function that satisfies the following Lipschitz condition:

$$\|f(k, \varphi) - f(k, \psi)\| \leq L_f(k)(\|\varphi(0) - \psi(0)\| + \dots + \|\varphi(-\mu) - \psi(-\mu)\|), \quad (3.8)$$

for all $\varphi, \psi \in \mathcal{B}$ and each $k \in \mathbb{Z}$, where $L_f: \mathbb{Z} \rightarrow \mathbb{R}^+$ is l^p -summable and $f(\cdot, 0) \in l^p(\mathbb{Z}, \mathbb{X})$. Then there is a unique bounded solution $u(n)$ of Eq. (1.1) such that $u \in l^p(\mathbb{Z}, \mathbb{X})$.

Proof. We continue to use the same notations as in the proof of Theorem 1.1. For the sake of brevity we only give a sketch of the proof. Let u be in $l^p(\mathbb{Z}, \mathbb{X})$, we have that

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \|\mathfrak{F}u(n)\|^p &\leq 2^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| L_f(j) \sum_{\tau=0}^{\mu} \|u(j-\tau)\| \right)^p + 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}+1} \sum_{n=-\infty}^{\infty} \|f(n, 0)\|^p \\ &\leq 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}+1} 2^{\mu p} \|L_f\|_{l^p(\mathbb{Z})}^p \|u\|_{l^p(\mathbb{Z}, \mathbb{X})}^p + 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}+1} \|f(\cdot, 0)\|_{l^p(\mathbb{Z}, \mathbb{X})}^p. \end{aligned}$$

Therefore we have that

$$\|\mathfrak{F}u\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq 2 \|s(\lambda, \cdot)\|_1 \left(2^\mu \|L_f\|_{l^p(\mathbb{Z})} \|u\|_{l^p(\mathbb{Z}, \mathbb{X})} + \|f(\cdot, 0)\|_{l^p(\mathbb{Z}, \mathbb{X})} \right).$$

Hence $\mathfrak{F}$ is well defined.

Let u and v be two functions in $l^p(\mathbb{Z}, \mathbb{X})$. We have that

$$\|\mathfrak{F}u - \mathfrak{F}v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{n=-\infty}^{\infty} L_f(n)^p \left(\sum_{\tau=0}^{\mu} \|u(n-\tau) - v(n-\tau)\| \right)^p \right)^{\frac{1}{p}} \leq \|s(\lambda, \cdot)\|_1 2^\mu \|L_f\|_{l^p(\mathbb{Z})} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

Since

$$\|\mathfrak{F}u(n) - \mathfrak{F}v(n)\| \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n-1} 2^\mu L_f(j)^p \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

We obtain that

$$\|\mathfrak{F}^{(2)}u - \mathfrak{F}^{(2)}v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \left(2^\mu \frac{a_\mu^2}{2} \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})},$$

where $a_\mu := \|s(\lambda, \cdot)\|_1 \|2^\mu L_f\|_{l^p(\mathbb{Z})}^p$. By induction we get

$$\|\mathfrak{F}^{(m)}u - \mathfrak{F}^{(m)}v\|_{l^p(\mathbb{Z}, \mathbb{X})} \leq \left(2^\mu \frac{a_\mu^m}{m!} \right)^{\frac{1}{p}} \|u - v\|_{l^p(\mathbb{Z}, \mathbb{X})}.$$

This completes the proof. $\square$

Example 3.2. Let ς be in $\mathbb{C}$ such that $|\varsigma| < 1$ and take $\lambda \in \mathbb{D}(-\varsigma, 1)$. Let $l: \mathbb{Z} \rightarrow \mathbb{R}^+$ be a function such that $\sum_{j=-\infty}^{\infty} l(j)^p < +\infty$. We consider the following Volterra functional difference equation in the Banach space $\mathbb{X}$:

$$u(n+1) = \lambda \sum_{j=-\infty}^n \varsigma^{n-j} u(j) + l(n) \left(\frac{u(n)}{\|u(n)\| + 1} + \frac{u(n-1)}{\|u(n-1)\| + 1} \right), \quad n \in \mathbb{Z}. \quad (3.9)$$

By Corollary 3.1 there is a unique l^p -bounded solution of (3.9).

3.2. Proof of Theorem 1.2

We denote by $\ell_m^p(\mathbb{Z}, \mathbb{X})$ the Banach space of all sequences $u \in \ell^p(\mathbb{Z}, \mathbb{X})$ such that $u(n) = 0$ for all $n \geq m + 1$ equipped with the norm $\|\cdot\|_{\ell^p(\mathbb{Z}, \mathbb{X})}$. For $r > 0$, denote $B_r(\ell_m^p(\mathbb{Z}, \mathbb{X}))$ the ball $\|u\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq r$ in $\ell_m^p(\mathbb{Z}, \mathbb{X})$.

Proof. of Theorem 1.2 Let β be in $(0, 1)$. Using (iii) and (ii) there are n_1 and n_2 in $\mathbb{Z}$ such that:

$$2\|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n_1} \|f(j, 0)\|^p \right)^{\frac{1}{p}} \leq \beta a$$

and

$$\mathcal{T} := \beta + 2\|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n_2} \ell^f(j, a)^p \right)^{\frac{1}{p}} < 1.$$

Take $m = \min\{n_1, n_2\}$. We define the operator $\mathcal{H} : B_a(\ell_m^p(\mathbb{Z}, \mathbb{X})) \rightarrow B_a(\ell_m^p(\mathbb{Z}, \mathbb{X}))$ by

$$\mathcal{H}u(n) = \begin{cases} 0, & \text{if } n \geq m + 1, \\ \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)f(j, u_j), & \text{if } n \leq m. \end{cases} \quad (3.10)$$

Let u be in $B_a(\ell_m^p(\mathbb{Z}, \mathbb{X}))$, we will prove that $\mathcal{H}u$ belongs to $B_a(\ell_m^p(\mathbb{Z}, \mathbb{X}))$. Let p and q be conjugated exponents. Proceeding in similar way like (3.3) and keeping in mind Remark 2.1 we get

$$\begin{aligned} \sum_{n=-\infty}^m \|\mathcal{H}u(n)\|^p &\leq 2^p \sum_{n=-\infty}^m \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, u_j) - f(j, 0)\| \right)^p + 2^p \sum_{n=-\infty}^m \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, 0)\| \right)^p \\ &\leq 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{n=-\infty}^m \sum_{j=-\infty}^{n-1} |s(\lambda, n-j)| \ell^f(j, \rho a)^p \|u(j)\|^p + 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{n=-\infty}^m \sum_{j=-\infty}^{n-1} |s(\lambda, n-j)| \|f(j, 0)\|^p \\ &\leq 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{j=-\infty}^{m-1} \sum_{n=j}^{m-1} |s(\lambda, n-j)| \ell^f(j, \rho a)^p \|u(j)\|^p + 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{j=-\infty}^{m-1} \sum_{n=j}^{m-1} |s(\lambda, n-j)| \|f(j, 0)\|^p \\ &\leq 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \left(\sum_{j=-\infty}^{n_2} \ell^f(j, \rho a)^p \right) \|u\|_{\ell^p(\mathbb{Z}, \mathbb{X})}^p + 2^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}+1} \left(\sum_{j=-\infty}^{n_1} \|f(j, 0)\|^p \right). \end{aligned}$$

We have thus proved

$$\|\mathcal{H}u\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq (1 - \beta)a + \beta a = a,$$

whence

$$\mathcal{H}u \in B_a(\ell_m^p(\mathbb{Z}, \mathbb{X})).$$

Let u and v be two functions in $B_a(\ell_m^p(\mathbb{Z}, \mathbb{X}))$, we have that

$$\begin{aligned} \|\mathcal{H}u - \mathcal{H}v\|_{\ell^p(\mathbb{Z}, \mathbb{X})} &\leq \left(\sum_{n=-\infty}^m \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \ell^f(j, \rho a)^p \|u(j) - v(j)\| \right)^p \right)^{\frac{1}{p}} \\ &\leq \|s(\lambda, \cdot)\|_1^{\frac{1}{q}} \left(\sum_{j=-\infty}^{m-1} \sum_{n=j}^{m-1} |s(\lambda, n-j)| \ell^f(j, \rho a)^p \|u(j) - v(j)\|^p \right)^{\frac{1}{p}} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{n_2} \ell^f(j, \rho a)^p \right)^{\frac{1}{p}} \|u - v\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \\ &\leq \left(\frac{\mathcal{T} - \beta}{2} \right) \|u - v\|_{\ell^p(\mathbb{Z}, \mathbb{X})}. \end{aligned}$$

Since $\frac{\mathcal{T} - \beta}{2} < \frac{1}{2}$, the operator $\mathcal{H}$ is a $(\frac{\mathcal{T} - \beta}{2})$ -contraction. This completes the proof of the theorem. $\square$

Example 3.3. Let $\gamma > 0$. We consider the phase space $\mathcal{B}^\gamma(\mathbb{X})$ defined in Example 2.1. Let ς be in $\mathbb{C}$ such that $|\varsigma| < 1$ and take $\lambda \in \mathbb{D}(-\varsigma, 1)$. Let g be a function in $\ell^p(\mathbb{Z})$. We consider the following difference equation in $\mathbb{X}$:

$$u(n+1) = \lambda \sum_{j=-\infty}^n \varsigma^{n-j} u(j) + g(n) \|u(n)\|^2 u(n), \quad n \in \mathbb{Z}. \quad (3.11)$$

Let $f(k, \varphi)$ be the perturbation associated to (3.11), then

$$\|f(k, \varphi) - f(k, \psi)\| \leq |g(k)| \left(\|\varphi\|_\gamma^2 + (\|\psi\|_\gamma + \|\varphi\|_\gamma) \|\psi\|_\gamma \right) \|\varphi(0) - \psi(0)\|, \quad \text{for all } \varphi, \psi \in \mathcal{B}^j(\mathbb{X}) \text{ and } k \in \mathbb{Z}.$$

We can choose the function $\ell^j(k, R)$ in condition $(\mathbf{H}_{loc})$ as $\ell^j(k, R) = 3|g(k)|R^2$. By Theorem 1.2 there is a constant $m \in \mathbb{Z}$ and a unique ℓ^p -solution $u(n)$ of (3.11) for all $n \leq m$.

Remark 3.1. Under similar conditions as Theorem 1.2 if we substitute condition (i) of $(\mathbf{H}_{loc})$ by the following condition.

(i)* For each positive number R , for all $k \in \mathbb{Z}$, and $\|\varphi\|_B \leq R, \|\psi\|_B \leq R$

$$\|f(k, \varphi) - f(k, \psi)\| \leq \ell^j(k, R)(\|\varphi(0) - \psi(0)\| + \dots + \|\varphi(-\mu) - \psi(-\mu)\|).$$

The same conclusion of the Theorem 1.2 is true. The proof of this assertion requires small modifications in the demonstration of the Theorem 1.2. The proof is left to the reader.

3.3. Proof of Theorem 1.3

Proof. of Theorem 1.3 We define the operator $\mathfrak{F} : \ell^p(\mathbb{Z}, \mathbb{C}) \rightarrow \ell^p(\mathbb{Z}, \mathbb{C})$ by (3.2). We divide the proof into several steps.

Step 1 The operator $\mathfrak{F}$ is well defined.

Let p and q be conjugated exponents and fix $u \in \ell^p(\mathbb{Z}, \mathbb{C})$, then

$$\sum_{n=-\infty}^{\infty} |(\mathfrak{F}u)(n)|^p \leq \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \mathcal{W}_f(j, \rho \|u\|_d) \right)^p \leq \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{j=-\infty}^{\infty} \left(\sum_{n=0}^{\infty} |s(\lambda, n)| \right) \mathcal{W}_f(j, \rho \|u\|_d)^p.$$

Therefore

$$\|\mathfrak{F}u\|_{\ell^p(\mathbb{Z}, \mathbb{C})} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, \rho \|u\|_d)^p \right)^{1/p}.$$

Hence $\mathfrak{F}$ is well defined.

Step 2 The operator $\mathfrak{F}$ is continuous from $\ell^p(\mathbb{Z}, \mathbb{C})$ into itself.

From (ii) in condition $(\mathbf{H}_{sca})$ for each $\varepsilon > 0$ there is $\delta > 0$ such that for all $\varphi, \psi \in \mathcal{B}, \|\varphi - \psi\|_B < \delta$ implies

$$\sum_{j=-\infty}^{\infty} |f(j, \varphi) - f(j, \psi)|^p < \left(\frac{\varepsilon}{\|s(\lambda, \cdot)\|_1} \right)^p.$$

For $u, v \in \ell^p(\mathbb{Z}, \mathbb{C})$ with $\|u - v\|_{\ell^p(\mathbb{Z}, \mathbb{C})} < \frac{\varepsilon}{\rho}$, where ρ is given in Remark 2.1, then $\|u_j - v_j\|_B < \delta$, for each j . Therefore $\|\mathfrak{F}u - \mathfrak{F}v\|_{\ell^p(\mathbb{Z}, \mathbb{C})} \leq \varepsilon$. In fact, it follows from the following estimate:

$$\sum_{n=-\infty}^{\infty} |(\mathfrak{F}u)(n) - (\mathfrak{F}v)(n)|^p \leq \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| |f(j, u_j) - f(j, v_j)| \right)^p \leq \|s(\lambda, \cdot)\|_1^{\frac{p}{q}+1} \left(\frac{\varepsilon}{\|s(\lambda, \cdot)\|_1} \right)^p.$$

Step 3 The operator $\mathfrak{F}$ is completely continuous.

Let r be a positive real number. Put $\mathcal{V} = \mathfrak{F}(B_r(\ell^p(\mathbb{Z}, \mathbb{C})))$.

(a) We observe that $\mathcal{V}$ is bounded. In fact, we can verify that $\sup_{u \in \mathcal{V}} \|\mathfrak{F}u\|_{\ell^p(\mathbb{Z}, \mathbb{C})}$ is dominated by

$$\|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, \rho r)^p \right)^{1/p}.$$

(b) We claim that $\sum_{i=n}^{\infty} |(\mathfrak{F}u)(i)|^p \rightarrow 0$ as $n \rightarrow \infty$ uniformly in $u \in B_r(\ell^p(\mathbb{Z}, \mathbb{C}))$.

We get

$$\sum_{i=n}^{\infty} |(\mathfrak{F}u)(i)|^p \leq \sum_{i=n}^{\infty} \left(\sum_{j=-\infty}^{i-1} |s(\lambda, i-1-j)| \mathcal{W}_f(j, \rho r) \right)^p \leq \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \sum_{i=n-1}^{\infty} \sum_{j=-\infty}^i |s(\lambda, i-j)| \mathcal{W}_f(j, \rho r)^p.$$

Since

$$\sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^i |s(\lambda, i-j)| \mathcal{W}_f(j, \rho r)^p = \|s(\lambda, \cdot)\|_1 \sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, \rho r)^p < \infty.$$

The last estimate guarantees the fact that $\lim_{n \rightarrow \infty} \sum_{i=n}^{\infty} |(\mathfrak{F}u)(i)|^p = 0$, uniformly in $u \in B_r(l^p(\mathbb{Z}, \mathbb{C}))$ and the claim is established. **(c)**

Analogously, we have that

$$\sum_{i=-\infty}^m |(\mathfrak{F}u)(i)|^p \leq \|s(\lambda, \cdot)\|_1^{\frac{p}{p+1}} \sum_{j=-\infty}^{m-1} \mathcal{W}_f(j, \rho r)^p,$$

whence $\lim_{m \rightarrow -\infty} \sum_{i=-\infty}^m |(\mathfrak{F}u)(i)|^p = 0$, uniformly in $u \in B_r(l^p(\mathbb{Z}, \mathbb{C}))$.

From **(a)**, **(b)**, **(c)** and using a compactness criterion in $l^p(\mathbb{Z}, \mathbb{C})$ given in Taylor's book [30, p. 286], we get that $\mathfrak{F}$ is completely continuous.

Step 4 We observe that $\mathfrak{F}(B_R(l^p(\mathbb{Z}, \mathbb{C}))) \subset B_R(l^p(\mathbb{Z}, \mathbb{C}))$, where R is given in condition **(H_{sca})**.

In fact, let u^R be in $B_R(l^p(\mathbb{Z}, \mathbb{C}))$ from (iii) in condition **(H_{sca})** we get

$$\|\mathfrak{F}(u^R)\|_{l^p(\mathbb{Z}, \mathbb{C})} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f(j, \rho R)^p \right)^{1/p} \leq R.$$

Applying now the Schauder's fixed point theorem, we infer that $\mathfrak{F}$ has a fixed-point u in $B_R(l^p(\mathbb{Z}, \mathbb{C}))$, which finishes the proof of Theorem 1.3. $\square$

Example 3.4. Let $\gamma > 0$. We consider the scalar phase space $\mathcal{B}'(\mathbb{C})$. Let ς be in $\mathbb{C}$ such that $|\varsigma| < 1$ and take $\lambda \in \mathbb{D}(-\varsigma, 1)$. Let $G : \mathbb{Z}^- \rightarrow \mathbb{C}$ and $F : \mathbb{Z} \rightarrow \mathbb{C}$ be two functions so that $F \in l^p(\mathbb{Z}, \mathbb{C})$, $p \geq 1$ and

$$\chi := \sum_{j=-\infty}^0 |G(j)| e^{-\gamma j} < \infty. \quad (3.12)$$

Let $h : \mathbb{C} \rightarrow \mathbb{C}$ be a uniformly continuous function. We assume in addition that there are constants $C > 0$ and $R > 0$, so that h satisfies $|h(z)| \leq C|z|^\beta$, $z \in \mathbb{C}$ with $\beta \in [0, 1]$ and

$$\frac{C}{R^{1-\beta}} \|s(\lambda, \cdot)\|_1 \chi^\beta \|F\|_{l^p(\mathbb{Z}, \mathbb{C})} < 1. \quad (3.13)$$

We consider the following scalar difference equation:

$$u(n+1) = \lambda \sum_{j=-\infty}^n \varsigma^{n-j} u(j) + F(n) h \left(\sum_{j=-\infty}^n G(j-n) u(j) \right), \quad n \in \mathbb{Z}, \quad (3.14)$$

with these definitions we can verify easily that the hypotheses of Theorem 1.3 hold. In fact, let $f(k, \varphi)$ be the perturbation associated to (3.14). It is not difficult to see that $|f(k, \varphi)| \leq C \chi^\beta |F(k)| \|\varphi\|_{l^p(\mathbb{Z}, \mathbb{C})}^\beta$, which (together with the fact that $F \in l^p(\mathbb{Z}, \mathbb{C})$) show that condition (i) in **(H_{sca})** holds with $\mathcal{W}_f(k, r) = C \chi^\beta |F(k)| r^\beta$. Taking into account that h is uniformly continuous it follows that the map $\varphi \in \mathcal{B}'(\mathbb{C}) \rightarrow f(\cdot, \varphi) \in l^p(\mathbb{Z}, \mathbb{C})$ is uniformly continuous. From (3.13) we have that condition (iii) in **(H_{sca})** is fulfilled. Consequently, by Theorem 1.3 we can assert that problem (3.14) has a l^p -bounded solution. It is worth to point out that we have obtained this result without assuming that h satisfies a local Lipschitz condition. This finishes the discussion of Example 3.4.

4. Our methods applied to other equations

To establish the next result, we need to introduce the following condition.

Condition (H_{met}): The following conditions hold:

(i) The function $f(k, \cdot) : l^p(\mathbb{Z}, \mathcal{B}) \rightarrow \mathbb{X}$ satisfies a Lipschitz conditions, that is for all $\xi, \eta \in l^p(\mathbb{Z}, \mathcal{B})$ and all $k \in \mathbb{Z}$, we get

$$\|f(k, \xi) - f(k, \eta)\| \leq \beta_f(k) \|\xi - \eta\|_{l^p(\mathbb{Z}, \mathcal{B})},$$

where $\beta_f := (\beta_f(k))$ is such that $\sum_{j=-\infty}^{\infty} \beta_f(j)^p < +\infty$.

(ii) $f(\cdot, 0) \in l^p(\mathbb{Z}, \mathbb{X})$.

Theorem 4.1. Let λ be in Ω_ς . Assume that $\mathcal{B}$ is a phase space that satisfies axiom **(B)** and that the condition **(H_{met})** holds. In addition suppose that the following condition holds

$$2\rho \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p \right)^{1/p} < 1, \quad (4.1)$$

where ρ is the constant given in Remark 2.1 (see Section 2.1). Then, there is a unique ℓ^p -bounded solution $u(n)$ of equation

$$u(n+1) = \lambda \sum_{j=-\infty}^n a(n-j)u(j) + f(n, E^0(u(\cdot))), \quad n \in \mathbb{Z}. \quad (4.2)$$

Moreover, one has the following a priori estimate for the solution $u(n)$ of (4.2)

$$\frac{J}{H} \|u\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq \|E^0(u(\cdot))\|_{\ell^p(\mathbb{Z}, \mathbb{B})} \leq \frac{2\rho \|s(\lambda, \cdot)\|_1 \|f(\cdot, 0)\|_{\ell^p(\mathbb{Z}, \mathbb{X})}}{1 - 2\rho \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p\right)^{1/p}},$$

where H and J are the constants of axiom (A) and

$$E^0(v)(t) = \begin{cases} v, & t = 0, \\ 0, & t < 0. \end{cases} \quad (4.3)$$

Proof. of Theorem 4.1 We define the operator $\mathcal{I} : \ell^p(\mathbb{Z}, \mathbb{B}) \rightarrow \ell^p(\mathbb{Z}, \mathbb{B})$ by

$$\mathcal{I}(\xi)(n) = \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)E^0(f(j, \xi)), \quad \text{where } \xi \in \ell^p(\mathbb{Z}, \mathbb{B}) \text{ and } n \in \mathbb{Z}. \quad (4.4)$$

Step 1 The operator $\mathcal{I}$ is well defined. Let ξ be a sequence in $\ell^p(\mathbb{Z}, \mathbb{B})$ and let p and q be conjugated exponents, we have see Remark 2.1

$$\begin{aligned} \sum_{n=-\infty}^{\infty} \|(\mathcal{I}u)(n)\|^p &\leq \rho^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \beta_f(j) \|\xi\|_{\ell^p(\mathbb{Z}, \mathbb{B})} + \sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, 0)\| \right)^p \\ &\leq (2\rho)^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \beta_f(j) \right)^p \|\xi\|_{\ell^p(\mathbb{Z}, \mathbb{B})}^p + (2\rho)^p \sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, 0)\| \right)^p \\ &\leq (2\rho)^p \|s(\lambda, \cdot)\|_1^{\frac{p}{q}} \left(\left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p \right) \|\xi\|_{\ell^p(\mathbb{Z}, \mathbb{B})}^p + \sum_{j=-\infty}^{\infty} \|f(j, 0)\|^p \right). \end{aligned}$$

Therefore $\mathcal{I}$ is well defined.

Step 2 We now show that the operator $\mathcal{I}$ has a unique fixed point in $\ell^p(\mathbb{Z}, \mathbb{B})$. Let ξ and η be two functions in $\ell^p(\mathbb{Z}, \mathbb{B})$. In view of condition (i) in $(\mathbf{H}_{\text{met}})$ we can estimate the ℓ^p -norm of $\mathcal{I}\xi - \mathcal{I}\eta$ by

$$\rho^p \left(\sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \beta_f(j) \right)^p \right)^{1/p} \|\xi - \eta\|_{\ell^p(\mathbb{Z}, \mathbb{B})} \leq \rho \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p \right)^{1/p} \|\xi - \eta\|_{\ell^p(\mathbb{Z}, \mathbb{B})}. \quad (4.5)$$

By (4.5) and the contraction principle, it follows that $\mathcal{I}$ has a unique fixed point $\xi \in \ell^p(\mathbb{Z}, \mathbb{B})$.

Step 3 Let $\xi \in \ell^p(\mathbb{Z}, \mathbb{B})$ be a fixed point of $\mathcal{I}$. Put $u(n) := \xi(n)(0)$. This step is devoted to a detailed examination of the fact that $u(n)$ is a solution of equation (4.2). We note that $u(n)$ satisfies the following identity

$$u(n+1) = \sum_{j=-\infty}^n s(\lambda, n-j)f(j, E^0(u(\cdot))), \quad n \in \mathbb{Z}.$$

We obtain that

$$\begin{aligned} \sum_{j=-\infty}^n a(n-j)\lambda u(j) &= \lambda \sum_{j=-\infty}^n a(n-j) \left(\sum_{\tau=-\infty}^{j-1} s(\lambda, j-1-\tau)f(\tau, E^0(u(\cdot))) \right) = \lambda \sum_{j=-\infty}^{n-1} \sum_{\tau=-\infty}^j a(n-1-j)s(\lambda, j-\tau)f(\tau, E^0(u(\cdot))) \\ &= \lambda \sum_{\tau=-\infty}^{n-1} \sum_{j=\tau}^{n-1} a(n-1-j)s(\lambda, j-\tau)f(\tau, E^0(u(\cdot))) = \sum_{\tau=-\infty}^{n-1} \left(\sum_{j=0}^{n-1-\tau} \lambda a(n-1-\tau-j)s(\lambda, j) \right) f(\tau, E^0(u(\cdot))) \\ &= \sum_{\tau=-\infty}^{n-1} s(\lambda, n-\tau)f(\tau, E^0(u(\cdot))) = \sum_{\tau=-\infty}^n s(\lambda, n-\tau)f(\tau, E^0(u(\cdot))) - s(\lambda, 0)f(n, E^0(u(\cdot))) \\ &= u(n+1) - f(n, E^0(u(\cdot))), \end{aligned}$$

which establishes that $u(\cdot)$ is the solution of Eq. (4.2).

Step 4 We now treat the uniqueness of solutions. Let $u = u(n)$ be a solution of (4.2) such that $u \in \ell^p(\mathbb{Z}, \mathbb{X})$, then $\xi := E^0(u(\cdot))$ belongs to $\ell^p(\mathbb{Z}, \mathcal{B})$. Put $x(n) := (\mathcal{I}\xi)(n)(0)$, so $x(n)$ is solution of (4.2), whence $z(n) = x(n) - u(n)$ satisfy

$$z(j+1) = \lambda \sum_{l=-\infty}^j a(j-l)z(l). \quad (4.6)$$

On the other hand, we have that

$$\begin{aligned} \sum_{j=-\infty}^n s(\lambda, n-j) \left[z(j+1) - \lambda \sum_{l=-\infty}^j a(j-l)z(l) \right] &= \sum_{j=-\infty}^n s(\lambda, n-j)z(j+1) - \lambda \sum_{j=-\infty}^n \sum_{l=-\infty}^j a(j-l)s(\lambda, n-j)z(l) \\ &= \sum_{l=-\infty}^{n+1} s(\lambda, n+1-l)z(l) - \lambda \sum_{l=-\infty}^n \sum_{j=l}^n a(j-l)s(\lambda, n-j)z(l) \\ &= \sum_{l=-\infty}^{n+1} s(\lambda, n+1-l)z(l) - \sum_{l=-\infty}^n \left(\lambda \sum_{j=0}^{n-l} a(n-l-j)s(\lambda, j) \right) z(l) \\ &= z(n+1) + \sum_{l=-\infty}^n s(\lambda, n+1-l)z(l) - \sum_{l=-\infty}^n s(\lambda, n+1-l)z(l) = z(n+1). \end{aligned} \quad (4.7)$$

From (4.6) and (4.7) we get $z(n) \equiv 0$, which implies the uniqueness of solutions of (4.2).

Step 5 A priori estimates

Let $\xi \in \ell^p(\mathbb{Z}, \mathcal{B})$ be the unique fixed point of $\mathcal{I}$. From condition $(\mathbf{H}_{\text{met}})$ and **Step 1**, we have

$$\begin{aligned} \|\xi\|_{\ell^p(\mathbb{Z}, \mathcal{B})} &= \left[\sum_{n=-\infty}^{\infty} \left\| \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)E^0(f(j, \xi)) \right\|_{\mathcal{B}}^p \right]^{1/p} \leq \rho \left[\sum_{n=-\infty}^{\infty} \left(\sum_{j=-\infty}^{n-1} |s(\lambda, n-1-j)| \|f(j, \xi)\| \right)^p \right]^{1/p} \\ &\leq 2\rho \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p \right)^{1/p} \|\xi\|_{\ell^p(\mathbb{Z}, \mathcal{B})} + 2\rho \|s(\lambda, \cdot)\|_1 \|f(\cdot, 0)\|_{\ell^p(\mathbb{Z}, \mathbb{X})}. \end{aligned}$$

Hence

$$\|\xi\|_{\ell^p(\mathbb{Z}, \mathcal{B})} \leq \frac{2\rho \|s(\lambda, \cdot)\|_1 \|f(\cdot, 0)\|_{\ell^p(\mathbb{Z}, \mathbb{X})}}{1 - 2\rho \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \beta_f(j)^p \right)^{1/p}}.$$

It is easy to see that $J\|\xi(\cdot)(0)\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq H\|\xi\|_{\ell^p(\mathbb{Z}, \mathcal{B})}$. This ends the proof of the theorem. $\square$

4.1. Local version of Theorem 4.1

To obtain the next result, we require that the following condition holds.

Condition $(\mathbf{H}_{\text{met}}^{\text{loc}})$: The following conditions hold.

(i) The function $f(k, \xi)$ is locally Lipschitz with respect to $\xi \in \ell^p(\mathbb{Z}, \mathcal{B})$, i.e. for each positive number R for all $k \in \mathbb{Z}$ and $\|\xi\|_{\ell^p(\mathbb{Z}, \mathcal{B})} \leq R$, $\|\eta\|_{\ell^p(\mathbb{Z}, \mathcal{B})} \leq R$

$$\|f(k, \xi) - f(k, \eta)\| \leq \beta_{\text{met}}(k, R) \|\xi - \eta\|_{\ell^p(\mathbb{Z}, \mathcal{B})},$$

where $\beta_{\text{met}} : \mathbb{Z} \times [0, \infty) \rightarrow [0, \infty)$ is a nondecreasing function with respect to the second variable.

(ii) There is a positive number a_{met} such that $\sum_{j=-\infty}^{\infty} \beta_{\text{met}}(j, a_{\text{met}})^p < +\infty$.

(iii) $f(\cdot, 0) \in \ell^p(\mathbb{Z}, \mathbb{X})$.

Theorem 4.2. Let λ be in Ω_s . Assume that $\mathcal{B}$ is a phase space that satisfies axiom $(\mathbf{B})$ and that the condition $(\mathbf{H}_{\text{met}}^{\text{loc}})$ holds. Then, there is a constant $m \in \mathbb{Z}$ and a unique ℓ^p -bounded solution $u(n)$ of (4.2) for all $n \leq m$ such that $u(n) = 0$ for all $n \geq m+1$ and $\|u\|_{\ell^p(\mathbb{Z}, \mathbb{X})} \leq a_{\text{met}}$.

4.2. Non-Lipschitzian case

We next study the existence of ℓ^p -solutions of (4.2) when the function f is not Lipschitz continuous. Let $\mathcal{B}_{\text{sca}}$ be a scalar phase space of functions defined from $\mathbb{Z}_-$ into $\mathbb{C}$. To establish our result we consider the following condition.

Condition (H_{sca}^*) : Let $f : \mathbb{Z} \times {}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca}) \rightarrow \mathbb{C}$ be a function such that the following conditions hold:

(i) There is a function $\mathcal{W}_f^* : \mathbb{Z} \times \mathbb{R}^+ \rightarrow \mathbb{R}^+$ nondecreasing with respect to the second variable with

$$\sum_{j=-\infty}^{\infty} \mathcal{W}_f^*(j, r)^p < \infty,$$

for all $r \geq 0$ so that $|f(k, \xi)| \leq \mathcal{W}_f^*(k, \|\xi\|_{{}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca})})$, for all $k \in \mathbb{Z}$ and $\xi \in {}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca})$.

(ii) The map $\xi \in {}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca}) \rightarrow f(\cdot, \xi) \in {}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})$ is uniformly continuous.

(iii) There is $R > 0$ such that $\frac{1}{R} \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f^*(j, \rho R)^p \right)^{1/p} < 1$.

Theorem 4.3. Let λ be in Ω_s . Assume that $\mathcal{B}_{sca}$ is a scalar phase space that satisfies axiom **(B)** and that the condition (H_{sca}^*) holds. Then, there is a ${}^I\mathbb{P}$ -bounded solution $u(n)$ of (4.2).

Proof. of Theorem 4.3 To reader's convenience, we will briefly outline the argument of the proof. We define the operator $\mathcal{A} : {}^I\mathbb{P}(\mathbb{Z}, \mathbb{C}) \rightarrow {}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})$ by

$$\mathcal{A}u(n) = \sum_{j=-\infty}^{n-1} s(\lambda, n-1-j)f(j, E^0(u(\cdot))), \quad (4.8)$$

where $u \in {}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})$ and $n \in \mathbb{Z}$.

Initially we check that $\mathcal{A}$ is well-defined. We observe that if $u \in {}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})$, then $E^0(u(\cdot)) \in {}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca})$. We have the following estimate:

$$\|\mathcal{A}u\|_{{}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})} \leq \|s(\lambda, \cdot)\|_1 \left(\sum_{j=-\infty}^{\infty} \mathcal{W}_f^*(j, \rho \|u\|_{{}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})})^p \right)^{1/p}$$

where ρ is the constant given in Remark 2.1.

The continuity of operator $\mathcal{A}$ is an immediate consequence of condition (ii) of (H_{sca}^*) and the inequality

$$\|E^0(u(\cdot)) - E^0(v(\cdot))\|_{{}^I\mathbb{P}(\mathbb{Z}, \mathcal{B}_{sca})} \leq \rho \|u - v\|_{{}^I\mathbb{P}(\mathbb{Z}, \mathbb{C})}.$$

The remains of the proof is essentially the same of Theorem 1.3 (see Steps 3 and 4 of its proof). $\square$

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